NUS-ISS · CERTIFICATE IN INTELLIGENT REASONING SYSTEMS · PRACTICE MODULE · PROJECT PROPOSAL

ED Triage Pre-Arrival Intake Assistant

A caregiver's chat that starts only after the decision to go to the emergency department, asks what a triage nurse would ask, and hands the nurse a one-page, page-cited summary before the patient is seen.

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1 Introduction

Patients often travel to an emergency department (ED) with a family member. The family member usually knows the patient's medications, allergies, and what happened — but nobody collects this information until the triage nurse asks for it on arrival.

We propose a chat assistant that runs on the family member's phone during the trip. It starts after the decision to go to the ED has been made, and it asks the questions a triage nurse would ask. On arrival, the nurse scans a QR code and sees a one-page summary.

Goals

- Give the triage nurse a complete AMPLE history (Allergies, Medications, Past history, Last meal, Events) before the patient is seen, where every item shows who reported it and when.
- Make every rule conclusion cite the page of the Emergency Severity Index (ESI) triage handbook it comes from.
- Show the data-learned view next to the ESI rules' view, so the nurse sees where they agree and where they differ.
- Measure each design choice against its alternative.

Why it matters

Triage is a fast decision made with the information at hand. The history is the slowest part: it is collected by voice, often from a worried relative. No product we surveyed collects it from the family member before arrival (section 2). Our system adds this information without changing any hospital procedure.

2 Project background and market context

The problem. Triage nurses assign an acuity level from chief complaint, vital signs and history in a few minutes. The parts of the history that change acuity — anticoagulants after a fall, immunosuppression with fever, pregnancy with abdominal pain — are exactly the ones a family member may not think to volunteer. Ambulance patients are pre-announced by paramedics; patients arriving by car or on foot have no equivalent channel.

Where the system sits. The hospital process runs exactly as today; our system runs beside it and only hands information to specific steps (Figure 1). The chat starts after the decision to seek emergency care and never advises on whether to go — which keeps the system outside regulated symptom-checker territory — and there is no connection into hospital IT: the QR code is a plain link with a random session id.



Figure 1. Where the system sits in the hospital process — blue = the hospital process, unchanged; green = our system. Solid arrows are gates (hospital steps that our steps are designed to follow); dashed arrows are assists (information handed to a step).

Landscape: existing products near this market. Five adjacent product classes exist; none collects clinically structured history from the caregiver between the decision to go and arrival.

CLASS	NAMED EXAMPLES	WHAT THEY DO — AND WHAT THEY LEAVE OPEN
Telehealth consultations	Doctor Anywhere (Singapore)	A short triage questionnaire is completed before the call "to help the doctor serve you better"; a licensed doctor then consults by video and may refer the patient onward (Doctor Anywhere, 2021). The questionnaire serves that one consultation — when a patient is directed to in-person care, no structured record travels with them.
Consumer symptom checkers	Ada Health; Symptomate (Infermedica); Mediktor; Clearstep	Advise on whether and where to seek care, before the decision; several are regulated medical-device software (Ada is CE-marked Class IIa). We deliberately operate after that decision and give no care advice.
ED online check-in and queue visibility	InQuicker; Experity (Clockwise.MD); in Singapore, the HealthHub and NUHS apps show ED waiting times	Reserve a spot, show waiting times, handle registration — administrative data, not clinical history.
Prehospital pre-notification	Pulsara; Twiage (TigerConnect EMS)	Paramedic-to-hospital alerting with vitals, ECG and arrival time — for ambulance cases only, operated by clinicians, not families.
In-ED patient communication	Vital ERAdvisor	Keeps the patient informed (wait times, lab status) after arrival; collects no history.

Nearest precedent. The NUHS app offers a **pre-arrival health declaration** for its EDs and urgent-care centre, which a caregiver may also complete on the patient's behalf (symptoms, travel history) and show to screeners on arrival (NUHS, 2025). This confirms demand for pre-arrival intake in Singapore. It is, however, a fixed form feeding registration: it cannot ask follow-up questions, check what it collects against triage criteria, or update after submission — the gap this project addresses.

3 Literature review and market research

The project is product-oriented: a short academic grounding, then market research.

3.1 Academic grounding

- **Triage scales as encodable knowledge.** Two candidate triage scales were considered:
  - **ESI** (Emergency Nurses Association, 2023) — a fully published four-decision-point algorithm; its fourth edition (Gilboy et al., 2012) contains worked practice cases against which an encoding can be tested.
  - **PACS** (P1–P4), used in Singapore — category descriptions applied with nurse judgement; no criteria-level rulebook is published.

  A Singapore study comparing the two on 300 actual patients found moderate inter-rater reliability for both (ESI 0.59, PACS 0.49) and that ESI, unlike PACS, discriminates patients' resource needs (Fong et al., 2019). We therefore encode ESI, the published and testable algorithm, and treat a PACS mapping as an extension.
- **Real triage data.** Hong et al. (2018) released 560,486 de-identified ED visits with triage vitals, chief-complaint categories, medications, history and the nurse-assigned ESI level. It supplies training data for the practice signal and a comparator for the ESI rules.
- **Neuro-symbolic architecture.** In Kautz's taxonomy (Kautz, 2022) the system is *Neuro | Symbolic* — a neural model converts free text into symbolic facts consumed by a symbolic reasoner — and *Symbolic[Neuro]* — a symbolic interview controller calls the neural model as a subroutine. Decision authority is symbolic throughout; the model never touches acuity.
- **Lay-to-clinical bridging.** Public patient-facing instruments (BE-FAST stroke signs, NIMH ASQ, AAOS limb warning signs) map a caregiver's words to the fields a criterion reads, one citation per mapping.
- **Relation to prior work.** Symptom checkers reason about where to seek care; machine-learned triage predictors learn nurse practice and present it as the answer. This project does neither: the ESI handbook's criteria are executed deterministically with citations, and the learned model's view is displayed separately, labelled, with its limits stated.

3.2 Market research

- **Primary user.** The triage nurse, for whom the one-page report is designed to be read in under a minute.
- **Secondary user.** The ED doctor, who can open the same report at consultation.
- **Data source.** The caregiver or patient answering on the phone; not the customer.

4 Project scope

In scope

1. **Knowledge engineering** — the ESI algorithm as a page-cited rulebook, interview-control rules, lay→clinical bridges, and a documented acquisition pipeline (section 5).
2. **Deterministic engine** — a condition evaluator, a question selector that chooses the next question from the facts still missing, and an ESI decision-tree evaluator that maintains a minimum urgency level, which can only increase.
3. **Interview app** — chat with two model calls per turn (phrase the question; extract the answer), provenance on every fact, QR ticket, staff view.
4. **Practice signal** — a classifier trained on the Yale visits, applied to the same facts the rules use, displayed on the report as a separately labelled assessment beside the ESI rules' assessment.
5. **Nurse report** — an AMPLE layout produced by fixed program logic, showing the two assessments with their provenance and one AI-written paragraph automatically checked against the stored facts.
6. **Experiments** — each design choice is evaluated against its alternative (section 6), compared against the references listed in section 5.

Out of scope

- Advice on whether to seek care.
- Hospital IT integration (FHIR is future work).
- Assigning the triage level — the nurse does that.

Constraints and limitations

- The dataset is from a US hospital system, so the practice signal describes US practice and is labelled as such; transferability to Singapore is a stated limitation.
- Health data falls under the PDPA; detailed compliance work is outside this project's scope.
- The system verifies no identity: the report shows the identity as declared, labelled, for the nurse's usual check.
- The team has no clinical expertise: every clinical statement in the system is a citation, not a judgement of ours.

Technique groups

COMPONENT	TECHNIQUE	GROUP	COURSE
Page-cited ESI rulebook, agenda rules, minimum-urgency-level semantics, provenance guards, deterministic engine	Knowledge-based reasoning, business rules	1 — decision automation	MR
Practice signal beside the ESI rules' assessment; divergence analysis of ESI rules versus recorded practice; learned-tree vs rulebook structural comparison	Supervised learning on 560k real visits; knowledge discovery	3 — knowledge discovery & data mining	MR / RS
Slot-filling interview (dialogue policy as inference over slot state), schema-bound extraction, lay→clinical bridges, clinician summary generation	Natural-language interface, cognitive dialogue design	4 — cognitive techniques	CGS

5 Data collection and preparation

Knowledge sources

- The ESI handbook, fifth edition (Emergency Nurses Association, 2023), and the fourth edition with its practice cases (Gilboy et al., 2012).
- The CDC field-triage guideline (Sasser et al., 2012) and the NICE head-injury guideline NG232 (NICE, 2023).
- The AMPLE / SBAR / OLDCARTS history-taking instruments.
- The lay patient-facing instruments in section 3.1.

Rules are extracted from these sources under a fixed schema; each rule carries its source page and the quoted span it operationalises, and every rule is reviewed by a team member against that page.

Datasets and references for evaluation

DATASET	SIZE	ACCESS	ROLE
Yale ED visits (Hong et al., 2018)	560,486 visits × 972 columns	public download	persona seeding; practice-signal training; comparison of the ESI rules with recorded practice
ESI practice cases (Gilboy et al., 2012)	81 worked cases	in the handbook	reference for encoding fidelity
Scripted personas	20–25 planned	authored within the team	reference for interview accuracy
MIMIC-IV-ED demo (Johnson et al., 2023)	100 patients, ED stays with outcomes	open access (PhysioNet)	development set for outcome checks; the full ≈ 425k-stay set needs per-person credentialing and stays optional

Challenges

- Chief-complaint labels record the reason for the visit, not examination findings, so each mapping to a rule fact is a documented judgment.
- Vital signs are missing for many of the sickest visits.
- The recorded acuity is practice, not truth.
- Access to a subject-matter expert (triage nurse, doctor or paramedic) is the open item; the fallback is a structured review sheet an expert can complete asynchronously.

6 System design

Figure 2 shows the system's three parts, each with one job:

- **Rule engine** — chooses every question and evaluates answers against page-cited ESI criteria; outputs a minimum urgency level, the **floor** ("at least ESI-2, pending vital signs"), which can rise but never fall.
- **Language model** — phrases the questions and turns free-text answers into structured facts; decides nothing.
- **Classifier (practice signal)** — the green practice-lane box in Figure 2; trained on real ED visits, its view is shown beside the floor and never changes it.



Figure 2. System architecture — blue = symbolic components (they decide), orange = the language model (it translates, never decides), green = the data-learned practice signal (a labelled second lane that never moves the floor).

Design principles

- The chat starts only after the decision to go to the ED, and the only advice it can ever give is to call 995 or alert staff.
- The program chooses every question, danger signs first; the language model never controls the conversation.
- Every fact records who said it and how certain it is (reported → probable → confirmed); rules with safety consequences use only confirmed facts.
- The engine outputs a minimum urgency level together with the rule, its cited page and its provenance class — never a final level, never reassurance.
- The learned model's view appears beside the ESI rules' view and never changes it.
- The report is built by fixed program logic, and every change is listed when the report is next opened.

Planned evaluation

METRIC	HOW IT IS DERIVED	COMPARED AGAINST
Encoding fidelity — did we translate the ESI handbook correctly?	ESI handbook v4 worked practice cases are run through our rules and the output compared with the book's answer (criteria unchanged between editions only).	ESI handbook v4 practice cases (Gilboy et al., 2012)
Extraction accuracy and consistency	Share of each persona's known facts captured correctly; variation when the same persona is replayed.	the facts scripted for each persona
Rules versus model-direct triage	The same cases with the language model judging urgency directly: agreement, variation across reruns, presence of a citation.	ESI handbook v4 practice cases
End to end: a model-run agentic loop versus our rule-constrained architecture	Each persona played through both systems — a language model that runs the interview and assigns a level on its own, and our program-controlled interview with the rule-computed floor. Scored against the expected floor, with question order, rerun variation and citations recorded; disagreements are attributed to a missing rule, a missing fact, or a model error.	the persona's expected floor and scripted facts
Two-assessment agreement	How often the floor and the classifier agree on visits not used for training, with confidence intervals; the classifier is trained accuracy-optimised and class-balanced.	recorded nurse levels, as comparator
Floor soundness	Share of assessable visits where the history-only floor exceeds the recorded nurse level — a bound on over-calling.	recorded nurse levels, as comparator

What the nurse sees

The report is one page, opened from the QR link: the declared identity, labelled as not verified; the minimum urgency level with the rule that produced it; and the AMPLE history, each item tagged with its source. Figure 3 shows an illustrative design mock-up; the delivered report will follow the design principles above.

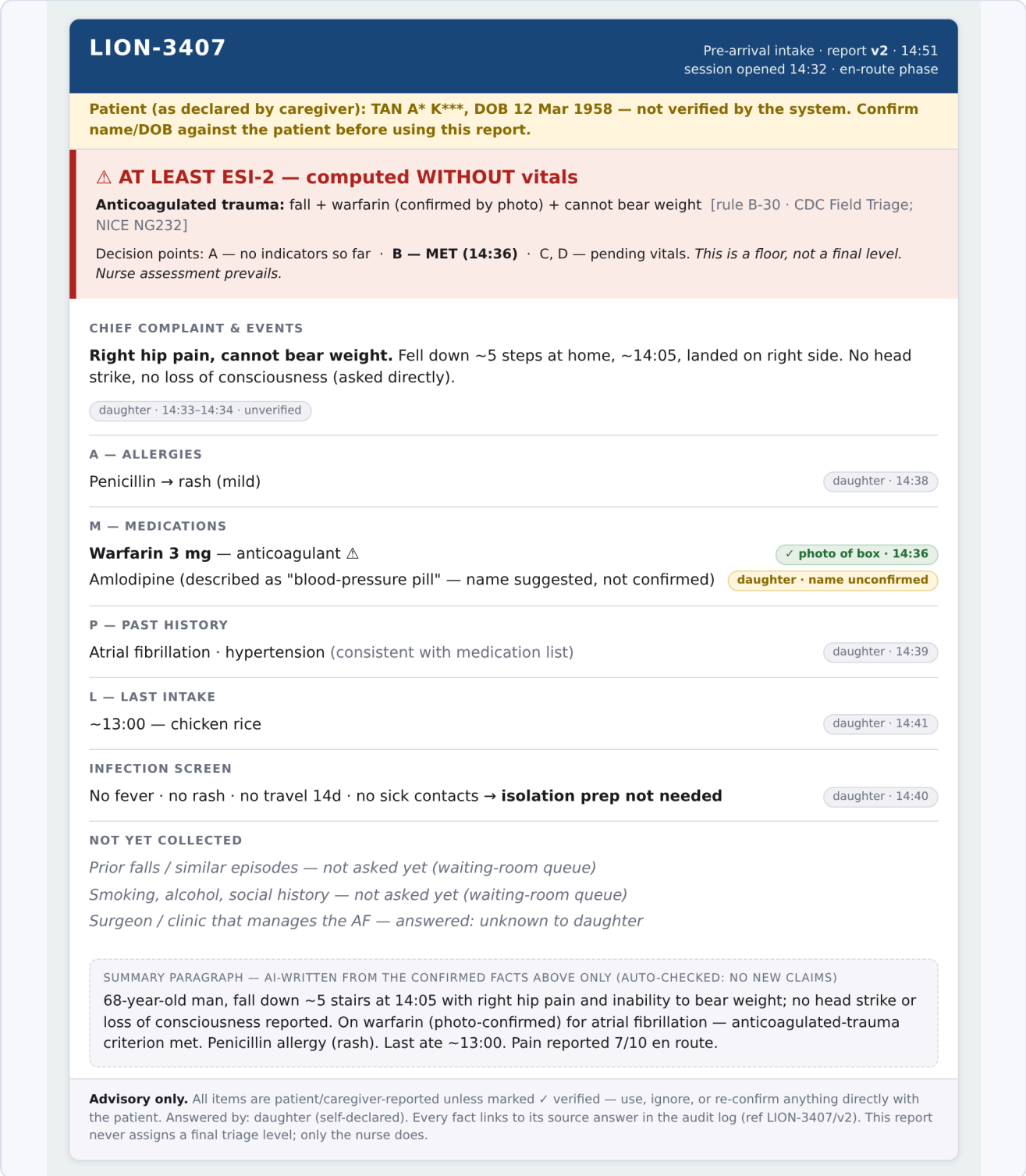


Figure 3. Design mock-up of the nurse's report at triage (illustrative).

7 Challenges and roadblocks

- **Verifying the rulebook.** Every machine-drafted rule must be checked by a person against its cited page; this is the largest single block of manual work.
- **Expert access.** The team has no clinical contact; without one, usefulness can be assessed only against published documents and data.
- **Lay language versus clinical criteria.** The ESI handbook's criteria assume clinical observation; mapping what a caregiver can say to what a rule reads is judgement work, and each mapping must itself cite a source.
- **Gaps and ambiguities in the source.** The handbook leaves some terms imprecise and in places states inconsistent thresholds; such cases need expert review or conservative handling.
- **Honest evaluation.** Scripted personas cannot reproduce everything real users do, so some qualities cannot be validated and will be reported as unvalidated.

8 Supplementary materials

Note on the use of AI. Language models are used (a) inside the system, for question phrasing and answer extraction only; (b) to draft rule entries from handbook text, each carrying a page citation and passing mechanical gates before a person verifies it; (c) as a coding and documentation assistant for the prototype and project documents, with team members reviewing every change.

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